***Situation 1:***

Suppose that you want to buy some vegetables and fruits that are given in the Fig. 7.20 with their weights and price and that you have a bag that capacity of carrying 15 kg. The objective is to buy more items within your budget of ₹550 and also that weight should not exceed 15 kg.



Fig. 7.20

You cannot buy all the items because if you calculate the total weight of all the items, then it would be greater than 15 kg (maximum capacity of your bag is 15 kg). So, let us try some approaches, to find how you can select more items so that you will buy them with maximum price within your budget of ₹550. For that let us tabulate the items with their weights and price you want to buy.

| Items |  |  |  |  |  |  |  |
|---|---|---|---|---|---|---|---|
| Weight (kg) | 1 | 3 | 5 | 4 | 1 | 3 | 2 |
| Price (₹) | 60 | 105 | 150 | 70 | 80 | 90 | 40 |

## Approach I - Selecting items with Maximum Price

In this approach, we select the items according to the maximum price. Here the maximum price in the table is ₹150/-. Now, let us tabulate to find the total price and how much can we buy vegetables and fruits within our budget and not exceeding 15kg.

| Items | Weight (kg) | Remaining weight to buy | Price (₹) |
|---|---|---|---|
|  | 5 | 15−5=10 | 150.00 |
|  | 3 | 10−3=7 | 105.00 |



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